Biomedical Signal Processing and Control 101 (2025) 107202

Contents lists available at ScienceDirect

Biomedical Signal Processing and Control

journal homepage: www.elsevier.com/locate/bspc

Edge-guided and hierarchical aggregation network for robust medical image segmentation

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ARTICLE INFO

Dataset link: https://github.com/tyjcbzd/EHANet

Keywords:
Medical image segmentation
Efficient fusion attention
Hierarchical feature aggregation
Inter-layer edge-aware module

ABSTRACT

Medical image segmentation with the convolutional neural networks (CNNs), significantly enhances clinical analysis and disease diagnosis. However, medical images inherently exhibit large intra-class variability, minimal inter-class differences, and substantial noise levels. Extracting robust contextual information and aggregating discriminative features for fine-grained segmentation remains a formidable task. Additionally, existing methods often struggle with producing accurate mask edges, leading to blurred boundaries and reduced segmentation precision. This paper introduces a novel Edge-guided and Hierarchical Aggregation Network (EHANet) which excels at capturing rich contextual information and preserving fine spatial details, addressing the critical issues of inaccurate mask edges and detail loss prevalent in current segmentation models. The Inter-layer Edge-aware Module (IEM) enhances edge prediction accuracy by fusing early encoder layers, ensuring precise edge delineation. The Efficient Fusion Attention Module (EFA) adaptively emphasizes critical spatial and channel features while filtering out redundancies, enhancing the model's perception and representation capabilities. The Adaptive Hierarchical Feature Aggregation Module (AHFA) module optimizes feature fusion within the decoder, maintaining essential information and improving reconstruction fidelity through hierarchical processing. Quantitative and qualitative experiments on four public datasets demonstrate the effectiveness of EHANet in achieving superior mIoU, mDice, and edge accuracy against eight other state-of-the-art segmentation methods, highlighting its robustness and precision in diverse clinical scenarios.

1. Introduction

Medical image segmentation is crucial for the accurate and automatic analysis of anatomical structures and pathological regions, playing a critical role in various clinical applications. The primary goal of this process is to partition medical images into specific subregions and extract regions of interest (ROIs) based on attributes like color, texture, and shape, thus laying the groundwork for subsequent tasks such as lesion region extraction, specific tissue measurements, and 3D reconstruction [1]. Over the past decades, object segmentation techniques in medical images have evolved significantly. Initially, manual segmentation by trained specialists was the standard. However, this method is prone to inconsistencies and lacks repeatability, making it unsuitable for managing large volumes of medical data [2]. Although traditional technology-based methods marked an improvement, they still faced challenges such as the need for setting appropriate thresholds in gray-scale threshold-based methods and selecting seed points in region-growth-based methods [3]. Thus, there is an urgent need to develop automated and accurate segmentation methods to provide a quantitative assessment for diagnosis, treatment planning, and monitoring of disease progression.

The advent of deep learning, particularly Convolutional Neural Networks (CNNs), has significantly advanced the field of medical image segmentation. U-Net and its variants [4–6], consisting of symmetric encoder and decoder architectures with skip connections, are extensively utilized to transfer features from the encoder path to the decoder path. A primary challenge in achieving accurate medical image segmentation is the guidance of boundary information. Due to the similarity in appearance between many pathological features and their surroundings, poorly defined boundaries frequently pose challenges. As depicted in the first two columns of Fig. 1, the distinction between the lesion region and the background is minimal, and the boundaries are unclear. In response, various methods have been developed to capture edge cues or extract edge-aware features to enhance segmentation performance. For instance, Wang et al. [7] describe a boundary-aware feature extractor that emphasizes edge detection and significantly improves segmentation accuracy near the boundaries. Similarly, Jiang et al. [8] import a dynamic contour adjustment mechanism that leverages feedback from

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https://doi.org/10.1016/j.bspc.2024.107202
Received 10 July 2024; Received in revised form 22 October 2024; Accepted 15 November 2024
Available online 26 November 2024
1746-8094/

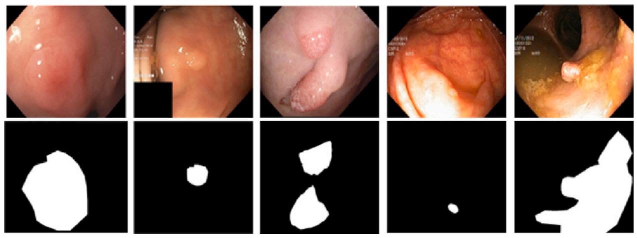

Fig. 1. Illustration of different challenges in Kvasir-SEG dataset, i.e., (1) The blurred edge between the lesion and its background (see the first two columns), and (2) The diversity of lesion scales (see the last three columns).

initial segmentation results to iteratively refine the contours of the segmented region. However, these methods often struggle to balance capturing detailed boundary information with maintaining the overall integrity of the segmented region. They may also exhibit limitations in complex situations where there is a high degree of variability in the size and shape of pathological features. Furthermore, the last three columns of Fig. 1 illustrate that the same target type can vary greatly in shape and size. Addressing how to effectively integrate features at different scales to accommodate the diversity of shapes, sizes, and locations of lesion regions poses another challenge. Among the current methods, Keidel et al. [9] utilize features at different scales to better suit lesion features of various sizes. Wu et al. [10] presents another method that employs a pyramid structure to modulate features at different levels, ensuring that both global and local contexts are adequately captured for enhanced segmentation of various lesions. Additionally, several studies have sought to overcome this challenge through different strategies, such as employing 3D fully convolutional networks stacked at multiple scales [1], context encoder networks, and multi-resolution residual connectivity networks to improve the discrimination of scale variations. While these approaches have made some progress, the challenge of effectively integrating multi-scale information without compromising spatial or semantic integrity remains.

In this paper, we introduce the Edge-guided and Hierarchical Aggregation Network (EHANet) for robust medical image segmentation. This network fully exploits multilevel and multiscale features, effectively captures contextual semantic information, and integrates edge-guidance to optimize segmentation accuracy and detail. Specifically, we introduce the Inter-layer Edge-aware Module (IEM), which refines edge prediction by integrating low-level and high-level features from different encoder layers, enhancing edge details by learning edge-attentive representations and effectively transferring them across layers. Efficient Fusion Attention Module (EFA) emphasizes critical spatial and channel features while filtering out irrelevant information. Further, the Adaptive Hierarchical Feature Aggregation Module (AHFA) ensures accurate reconstruction of segmented images by optimizing feature fusion within the decoder.

Our main contributions are three-fold:

- We fully utilize multi-level and multi-scale features to capture contextual semantic information and multi-scale characteristics, significantly boosting segmentation performance.
- The IEM enhances edge prediction by learning edge-attentive representations across early encoder layers. The EFA module adaptively emphasizes critical features, improving the model's representation capabilities. The AHFA module optimizes feature fusion within the decoder to maintain reconstruction fidelity and enhance expressive capability.
- We conducted experiments on four benchmark medical image segmentation tasks, and the results demonstrate that our proposed model outperforms state-of-the-art segmentation methods.

The remainder of this paper is organized as follows: Section 2 discusses works related to our study. Section 3 details the proposed medical segmentation network. Experimental comparisons to validate the effectiveness of our segmentation method are presented in Section 4. Finally, conclusions are drawn in Section 5.

2. Related work

2.1. Medical image segmentation

In the realm of computer-aided detection, medical image segmentation technology is crucial. Traditional segmentation methods include threshold-based, edge-based, region-based, and machine learning-based approaches [11]. Threshold-based methods identify foreground regions in a grayscale image by comparing pixel values against a predefined threshold. Edge-based methods delineate image segments using such features like changes in brightness level in the image, while region-based methods divide the image into multiple areas, assessing regions based on the similarity of features such as color, texture, and shape. Machine learning techniques employ a more comprehensive utilization of extracted features (including color, texture, and shape) and classify them using the support vector machines and pixel clustering methods [12]. However, traditional medical image segmentation methods often rely excessively on manual feature extraction, which is not only non-generalizable and complex but also time-consuming, making it challenging to meet clinical demands in terms of segmentation speed and accuracy.

Deep learning-based medical image segmentation methods have shown superiority over traditional methods in noise processing, segmentation accuracy, processing speed, and generalization ability. With the evolution of CNNs, researchers have developed various deep learning algorithms for medical image segmentation. Ronneberger et al. [4] introduce a novel deep learning algorithm known as U-Net. It efficiently retains an extensive level of detail and spatial information through skip connections, thereby enhancing network performance. Subsequently, several variants of the U-Net architecture have been developed. For instance, ResUNet++ [6] adds extra layers to ResUNet [5] to further enhance model performance. Additionally, Zhou et al. [13] densely connected skip connections to overlay and integrate features from different layers, compensating for information loss caused by downsampling and thus improving accuracy. Another innovative approach, MultiResUNet [14], replaces the traditional dual convolutional layer structure of the original U-Net with MultiRes blocks to diminish the semantic gap by adding additional convolutional units, thereby addressing generalization challenges. Furthermore, Butoi et al. [15] leverages cutting-edge deep learning techniques to ensure robust, accurate segmentation across various medical imaging modalities. Despite the existence of numerous medical image segmentation methods and their significant improvements in performance, effectively capturing high-level semantic features and adaptively extracting multi-scale information continue to pose challenges.

2.2. Edge-based methods

Current methods [13–15] in medical image segmentation are renowned for their efficient feature learning capabilities and precise segmentation of anatomical structures. However, these methods predominantly concentrate on enhanced feature learning and do not fully exploit boundary information. To tackle this issue, researchers have developed various strategies to incorporate boundary information effectively. Edge-based methods improve feature representation by integrating edge information within the network, which in turn boosts the overall performance of the model [8]. For instance, Chen et al. [16] introduced the BE-Net, a network designed to explicitly explore object boundaries from training images, ensuring the overlap of

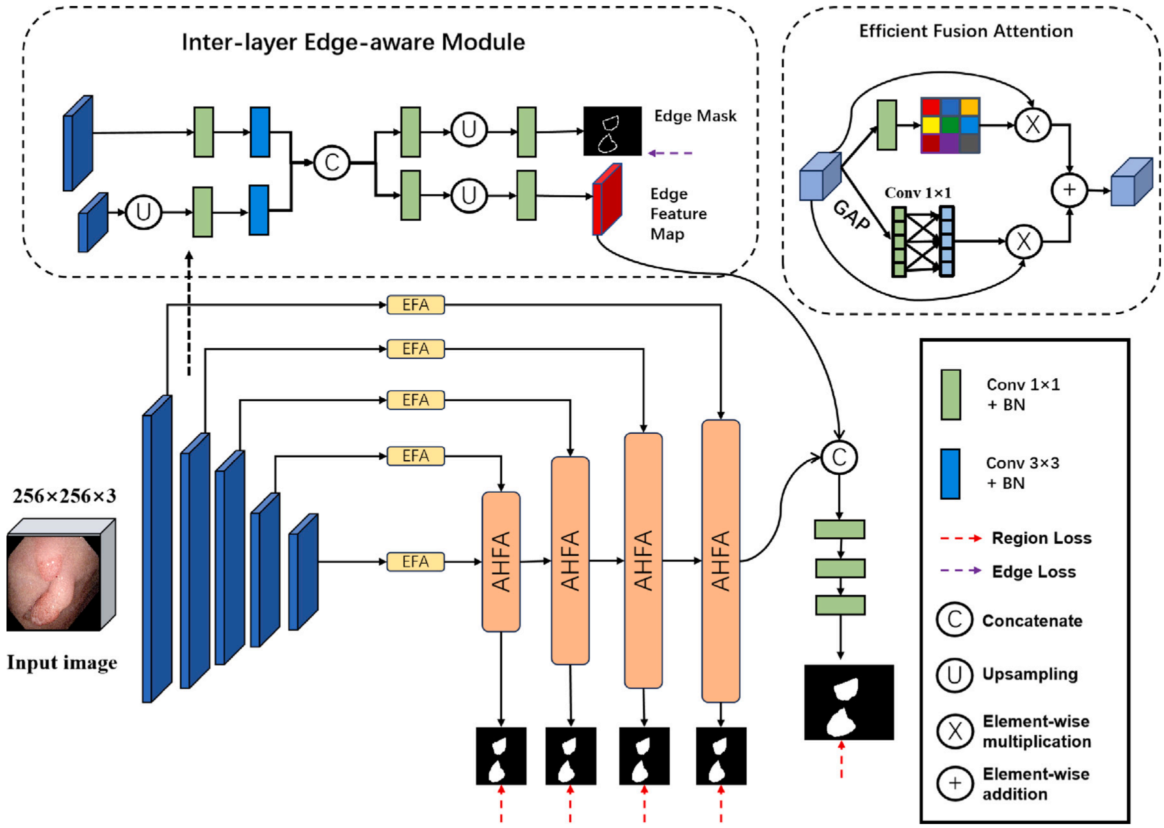

Fig. 2. Network structure of proposed Edge-guided and Hierarchical Aggregation Network (EHANet), consisting of three key components, i.e., inter-layer edge-aware module (Section 3.2), efficient fusion attention module (Section 3.3) and adaptive hierarchical feature aggregation module (Section 3.4).

segmentation and boundaries is maintained. Similarly, Ding et al. [17] present a method called Boundary-aware Feature Propagation, where edge information is merged into the feature propagation process to enhance segmentation accuracy in complex scenes. Furthermore, a generalized healthcare segmentation framework was proposed in [18], which employs an edge-attention mechanism to guide the segmentation process using edge-attention features. Additionally, an innovative and effective iterative edge attention network was introduced in [19]. Also, Fan et al. [11] developed a parallel inverse attention network to enhance the accuracy of polyp segmentation by leveraging edge features. Building on this, Wu et al. [10] proposed a boundary-guided module designed to learn boundary-enhanced feature representations, capturing both local features and boundary details to improve the detection of camouflaged objects. Meanwhile, a novel selective feature aggregation network with region and boundary constraints was proposed in [12] to predict polyp regions and boundaries more accurately. In contrast, EHANet adopts a different strategy. By combining edge-guided features with multi-scale features at the final stage of the network. This selective integration of edge information helps EHANet overcome the limitations of earlier edge-based methods, where edge information might be misaligned with the learned features.

2.3. Multi-scale fusion methods

Multi-scale fusion is a technique that merges information from multiple scales and various levels of abstraction, enhancing the performance of computer vision tasks such as segmentation, target detection, and image recognition. This approach has been widely adopted in various applications, including medical image segmentation, to capture regional contextual information and address scale variations. For instance, [20] introduced the Cross-Level Fusion and Contextual Reasoning Network, which employs cross-level fusion techniques to amalgamate information from different abstraction levels within the network architecture. Similarly, [21] developed the Multiscale Residual Fusion Network, specifically designed for medical image segmentation. This network improves the propagation of multiscale features and information flow through its Dual Scale Dense Fusion blocks, resulting in more accurate segmentation outcomes. Then, [22] unveiled Hierarchical Multiscale Fusion Network, which incorporates an axial multilayer perception and a Hierarchical Multiscale Fusion module to enhance segmentation accuracy in regions with small lesions and multiple loci. Additionally, [23] proposed the generalized multiscale subtraction network to address challenges such as inaccurate lesion localization and blurred lesion edges in medical image segmentation. Despite the significant advancements in multiscale and cross-level feature fusion methods, numerous opportunities remain for further enhancing segmentation performance. Unlike existing methods that focus on static multi-scale fusion, EHANet adaptively aggregates features across hierarchical levels, allowing it to dynamically adjust to the varying scales and complexities of medical images.

3. Method

In this section, we first give an overview of the proposed medical segmentation framework in Section 3.1. Then we introduce the three key components: inter-layer edge-aware module (Section 3.2), efficient fusion attention module (Section 3.3) and adaptive hierarchical feature aggregation module (Section 3.4). Finally, we introduce loss function in Section 3.5.

3.1. Overview

As depicted in Fig. 2, our EHANet employs an encoder–decoder architecture. The MobileViT-S [24] serves as the backbone of the encoder, where an input image $I \in \mathbf{R}^{W \times H \times 3}$ is first processed to extract multi-level features denoted as E_i (i = 1, 2, …, 5). Specifically, the

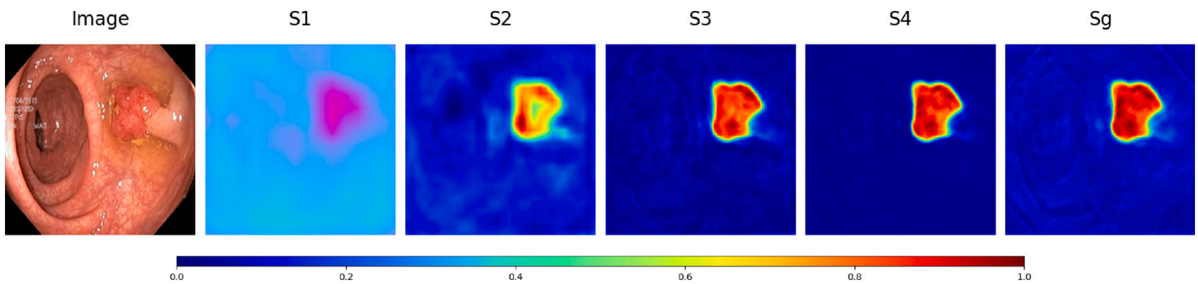

Fig. 3. Feature visualization results of proposed EHANet. Here, S1, S2, S3, S4 represent predicted masks from four cascade decoders. S_g represents the final predicted mask.

feature resolution of $\frac{W}{2} \times \frac{H}{2}$ for the first level, and a general resolution of $\frac{W}{2^i} \times \frac{H}{2^i}$ (i > 1) are obtained in this case. The number of channels for feature maps at the ith layer is given as C_i (i = 1, 2, …, 5), and we have C = [32, 64, 96, 128, 160]. As mentioned in [18], low-level features typically encapsulate intricate details such as edges and textures while high-level features embody rich semantic information that aids in the identification and segmentation of crucial object components. Consequently, low-level features E_1 and E_2 are directed to the inter-layer edge-aware module (IEM), which generates final edge mask and edge feature map. Given the challenges associated with redundant information in features directly concatenated via skip connections, which can mislead the network, it is crucial to emphasize key features initially [25]. Thus, the extracted features E_i are fed into EFA module, effectively filtering out non-essential elements. The decoder path is formed from four cascaded AHFA modules, which are used to preserve and enhance the distinct characteristics of the advanced refined features captured in the encoders and produce multi-scale segmentation maps (i.e. S_i, i = 1, 2, 3, 4).

In detail-rich and structurally complex image segmentation, accurate capture and representation of edges is crucial for improving the overall segmentation accuracy. By incorporating the edge information directly into the model, the model can be able to recognize and process the edge regions in the image more acutely, significantly improving the segmentation accuracy. Specifically, we integrate edge feature map F^e extracted by IEM with features from last decoder D_4. Finally, the module deeply processes the fused and optimized features through three consecutive 1×1 convolutional layers, and outputs the final mask result S_g.

To facilitate the intuitive visual comparison, we present several visual feature maps of the proposed network in Fig. 3. Owing to the limited space, we only provide some feature maps. Within the decoding branch, it is evident that detailed features, such as edges and textures, progressively become more distinct. Consequently, the integrated features extracted by the proposed network are both comprehensive and discriminative. This way further improves the interpretability of our model and helps radiologists understand the difference between diseased and normal tissues in clinical diagnostic analysis. Subsequently, we will elaborate on the superiority and implementation details of the three key modules.

3.2. Inter-layer edge-aware module

Several studies have demonstrated that edge information significantly enhances segmentation and detection performance [18,26]. Recognizing that low-level features preserve essential edge details, many existing methods incorporate these features to foster edge-enhanced representations. In this study, we introduce IEM, designed to substantially improve the richness and accuracy of edge predictions. The specific structure of the IEM is depicted in upper left in Fig. 2. The first layer of the encoder E_1 captures high-resolution low-level features, while the second layer E_2 processes lower-resolution, high-level features. Initially, IEM adjusts the second layer to match the resolution of the first layer via an upsampling operation, allowing for effective feature fusion. This adjustment is followed by a series of 1×1 and 3×3 convolution operations on these two feature streams. The 1×1 convolutions primarily perform feature reduction and selection to streamline computation while retaining critical edge information. Conversely, the 3×3 convolutions expand the receptive field, enhancing the network's ability to integrate broader spatial context information, thereby improving feature representation's spatial sensitivity. After feature fusion using concatenate, the IEM diverges into two pathways: one branch processes features through a sequence of 1×1 convolution, upsampling, and another 1×1 convolution. This path aims to refine features for providing detailed edge feature map which transmitted for predicting final mask S_g. The other branch focuses directly on edge detection, further processing the merged features through a 1×1 convolution and upsampling, followed by another 1×1 convolution, culminating with a softmax layer to predict the edge mask denoted as S_e.

3.3. Efficient fusion attention module

As depicted in upper right in Fig. 2, the EFA module adaptively learning the importance of different channels and spatial locations, which significantly improves the model's perception and representation capabilities, thereby enhancing performance in various image analysis and understanding tasks. Firstly, the spatial attention branch aims to enhance the model's ability to focus on the spatial location within the encoder features E_i. It captures the significance of different spatial locations by applying a 1×1 convolution. Subsequently, the convolved results are transformed into attention weights ranging from 0 to 1 using a sigmoid function. Ultimately, these weights are multiplied by the input to emphasize task-relevant spatial features while attenuating task-irrelevant spatial locations. The spatial-purified features can be denoted as f_i^s, and this process is depicted by

$$f_i^s = E_i \otimes \partial\left(\psi\left(E_i\right)\right) \tag{1}$$

where '∂' denotes sigmoid activation function, '$\otimes$' denotes element-wise multiplication, 'ψ' denotes 1×1 convolutional layer respectively.

Secondly, the channel attention branch extracts channel features from encoder features. It captures the significance of different channels by performing a global average pooling(GAP) operation, followed by a convolution operation to derive attention weights for each channel location. Similar to the spatial attention branch, these weights are also mapped by a sigmoid function. Further, these weights are multiplied by the input to emphasize task-relevant channel information and suppress task-irrelevant channels in the encoder features. The channel-purified features can be calculated by

$$f_i^c = E_i \otimes \partial\left(\psi\left(GAP\left(E_i\right)\right)\right) \tag{2}$$

where '*GAP*' denotes global average pooling operation.

The outputs from both the channel and spatial attention branches are then combined through element-wise addition. And we have

$$f_i^p = f_i^s \oplus f_i^c \tag{3}$$

where '$\oplus$' denotes element-wise addition.

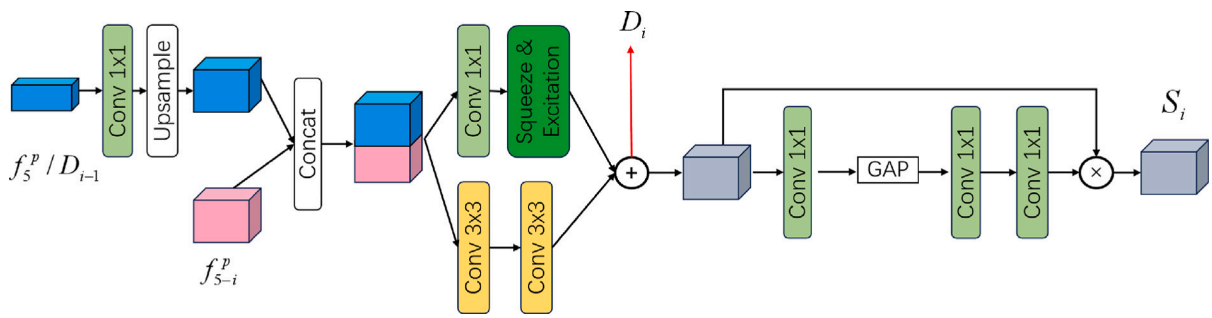

Fig. 4. The structure of Adaptive Hierarchical Feature Aggregation Module.

3.4. Adaptive hierarchical feature aggregation module

In deep neural networks, the encoder plays a vital role by extracting features from the original data. Conversely, the decoder is tasked with reconstructing these features to represent the original data accurately. However, many existing methods have focused on designing improved encoders to extract better features [5,6]. Nevertheless, these approaches often overlook the significance of the decoder, which directly contributes to the prediction process. This oversight can impede the efficient extraction and utilization of vital information from the feature maps. To address this gap, we propose AHFA module, specifically designed to improve the feature representation and expression capabilities of deep neural networks during the feature processing stage. The AHFA module is engineered to optimize feature fusion within the decoder, ensuring that the reconstructed output more accurately reflects the original input data. The structure of the module is illustrated in Fig. 4. Except the first AHFA module receives f_5^p and f_4^p as input features, other AHFA modules receive f_{5-i}^p from same level and features from previous AHFA module D_{i-1}. Specifically, input feature f_5^p or D_{i-1} transform to the same size of features f_4^p or f_{5-i}^p by 1×1 convolution and upsampling operations. Subsequently, the transformed features are fused with corresponding features f_4^p or f_{5-i}^p by using concatenate operation. Thus, we can obtain features F_i^{cat} by

$$F_i^{\text{cat}} = \begin{cases} C\left(up\left(\psi\left(f_5^p\right)\right), f_4^p\right) & i=1 \\ C\left(up\left(\psi\left(D_{i-1}\right)\right), f_{5-i}^p\right) & i=2,3,4 \end{cases} \tag{4}$$

where 'C' and 'up' denotes concatenate and upsampling operation.

Then, the fused feature representation F_i^{cat} enters two parallel branches. The first branch comprises two 3×3 convolutional layers, each followed by batch normalization and ReLU activation function. This branch is designed to extract higher-level feature representations through convolution operations, enabling the model to capture more complex and abstract features. The second branch initially employs a 1×1 convolutional layer and batch normalization layer to reduce feature dimensionality and computational complexity. Then, the attention mechanism is implemented by introducing the Squeeze Excitation layer. This layer enables adaptive feature weighting, allowing the model to emphasize important features and enhance their expressive capability. Finally, the outputs of the two branches are added and mapped through a sigmoid function to obtain D_i which will be used for upper AHFA module. The process can be denoted as

$$D_i = SE\left(\psi\left(F_i^{cat}\right)\right) \oplus \tau\left(\tau\left(F_i^{cat}\right)\right) \tag{5}$$

where 'SE' denotes Squeeze & Excitation layer, 'τ' denotes 3×3 convolutional layer.

Contrary to many existing methods that directly utilize the output from the highest decoder layer for prediction, thereby overlooking scale-correlated information during the restoration process. Our method capitalizes on the hierarchical nature of the decoding process, which allows us to derive region predictions based on the refined features captured at each decoding stage. Concretely, features first undergoes transformation via a 1×1 convolutional layer. This reduces the number of channels while retaining essential information. Subsequently, global contextual information D_i^p is extracted through GAP, which is

$$D_i^p = GAP\left(\psi\left(D_i\right)\right) \tag{6}$$

The extracted context is further refined through two consecutive 1×1 convolutions, aimed at enhancing feature quality and adjusting the channel-wise importance. The adaptive recalibration of the original features is facilitated by a sigmoid activation function, which scales the weights obtained from the second 1×1 convolution. These weights are then element-wise multiplied with the refined features to emphasize the model's focus on the key aspects of each feature map to produce refined region map of each decoder . The process is implemented by

$$S_i = D_i \otimes \psi\left(\psi\left(D_i^p\right)\right) \tag{7}$$

3.5. Loss function

In this study, we use $\mathcal{L}_{seg} = \mathcal{L}_r + \mathcal{L}_e$ as our segmentation loss function, where $\mathcal{L}_r$ denotes the binary cross-entropy loss (BCE), $\mathcal{L}_e$ is computed using the Lovasz-SoftMax method. The hierarchical predicted region masks S_i are up-sampled (denoted as S_i^{up}, i = 1, 2, 3, 4) to the same size of ground truth region mask G and final predicted region mask S_g. The predicted edge mask from IEM and ground truth region mask denote S_e and G^e, respectively. Generally, our overall segmentation loss function is given as

$$\mathcal{L}_{seg} = \mathcal{L}_r\left(S_g, G\right) + \sum_{i=1}^{4} \mathcal{L}_r\left(S_i^{up}, G\right) + \mathcal{L}_e\left(S_e, G^e\right) \tag{8}$$

Finally, we also provide the detailed algorithm training procedure is summarized below in Algorithm 1.

4. Experiments

In this section, we evaluate the performance of our proposed model against state-of-the-art (SOTA) models in medical image segmentation across four publicly available datasets. Initially, we describe the datasets and outline the employed metrics to evaluate the model's efficacy (Section 4.1 and Section 4.2, respectively). Detailed explanations regarding the training strategies and implementation specifics are provided in Section 4.3. Then, we compare the proposed model with the SOTA medical image segmentation methods in Section 4.4. To clarify the validity of the key parts in our model, we conduct ablation experiments in Section 4.5. Finally, we offer a thorough discussion of these findings in Section 4.6.

4.1. Datasets description

To thoroughly evaluate the proposed EHANet model, we conducted training and validation on four publicly available medical image datasets, each representing distinct imaging environments and challenges. This diversity tests the model's generalization capabilities across different conditions. The datasets are described below:

Algorithm 1 Training for EHANet

Input: Original Image Set $\{X_1, \dots, X_n\}$,
Region Label (ground truth) Set $\{G_1, \dots, G_n\}$,
Edge Label (ground truth) $\{G_1^e, \dots, G_n^e\}$
Output: Segmentation Region Map $\sum_{i=1}^{4} S_i$, S_g
Segmentation Edge Map S_e
Model parameters Θ
while not converge **do**
Sample X_i, G_i, G_i^e from $\{X_1, \dots, X_n\}$, $\{G_1, \dots, G_n\}$, $\{G_1^e, \dots, G_n^e\}$
Acquire different level feature map $\mathbf{E}_1, \dots, \mathbf{E}_5 = \text{Backbone}(X_i)$
Predict edge map and learn edge features $S^e, F^e = \text{IEM}(\mathbf{E}_1, \mathbf{E}_2)$
// Channel and spatial attention
$f_i^p = \text{EFA}(\mathbf{E}_1, \dots, \mathbf{E}_5)$
// Staged decoder prediction
$\sum_{i=1}^{4} S_i, D_4 = AHFA(f_i^p)$
// Edge-Guided Prediction
$S_g = Predict(F^e, D_4)$
// Total Loss Function
$\mathcal{L}_{seg} = \mathcal{L}_r\left(S_g, G\right) + \sum_{i=1}^{4} \mathcal{L}_r\left(S_i^{up}, G\right) + \mathcal{L}_e\left(S_e, G^e\right)$ as in Eq. (8)
The Adam optimizer and the total loss $\mathcal{L}_{seg}$ to update the network parameters Θ
end while

- **Kvasir-SEG** [27]: This open-access dataset comprises gastrointestinal polyp images and corresponding segmentation masks, manually annotated and verified by experienced gastroenterologists. It contains 1000 polyp images, with resolutions ranging from 332 × 487 to 1920 × 1072 pixels.
- **2018 Data Science Bowl** [28]: This dataset includes 670 images featuring a variety of cell types, including cancer cells and normal cells, thus providing a comprehensive set of cell image data for segmentation analysis.
- **ISIC 2018** [29]: Provided by the International Skin Imaging Collaboration (ISIC), this dataset offers a collection of skin lesion images with corresponding segmentation masks, manually labeled and validated by dermatologists. It encompasses approximately 10,000 images of various skin diseases and lesions, including melanoma, squamous cell carcinoma, basal cell carcinoma, among others.
- **LGG Datasets** [30]: Sourced from the National Cancer Institute's Cancer Imaging Archive, this dataset focuses on low-grade glioma (LGG) studies and features FLAIR modality images from TCGA patients. The abnormal regions in these images were manually labeled by neuroradiology professionals to support automated segmentation algorithm training.

All images were standardized by resizing them to a uniform scale to facilitate model training and evaluation. The preprocessed data was then divided into separate training, validation, and test datasets in a ratio of 8:1:1.

4.2. Evaluation metrics

In our research, inspired by [26], for general segmentation accuracy, we adopt two widely used metrics, including mean Dice score (mDice) and mean intersection over union (mIoU), which effectively measure the overlap accuracy between predicted segmentation and ground truth. Four metrics which are widely used in the field of object detection are introduced, including S-measure (S_α) [31], F-measure [32] (F_ω), E_φ [33], and mean absolute error (M). These metrics are pivotal for assessing the quality of object boundaries and detection accuracy. Additionally, we employ distance-based metrics to evaluate spatial discrepancies between the predicted and actual contours. These include the Average Symmetric Distance (Asd) and the Hausdorff Distance (Hd) [34]. Asd calculates the mean vertical distance across corresponding points on predicted and ground truth contours, offering a direct measure of spatial accuracy. Conversely, Hd assesses the maximum distance discrepancy between the most distant points of the predicted and actual contours, with lower values indicating higher precision in matching the true contour shape and position.

4.3. Training strategies and implementation details

In the realm of deep learning, researchers typically rely on large datasets to mitigate issues such as overfitting and data imbalance. Given the constraints in the volume of available medical training data, we utilized data augmentation techniques such as random cropping and rotation at [90°, 180°, 270°] to generate new datasets. This approach enhances the robustness of the trained model. However, data augmentation was not applied to the test set. Furthermore, all images were resized to a uniform resolution of 256 × 256 pixels before being input into the network for training. The proposed EHANet was implemented using PyTorch on a single NVIDIA 4060 Ti GPU with 16 GB of memory. We selected MobileVit-S [24] as the backbone network, which was pretrained on the ImageNet dataset [35]. For optimization, we employed the Adaptive Moment Estimation (Adam) optimizer, with an initial learning rate of $1e^{-4}$. The training process was conducted over a maximum of 200 epochs, with a batch size of 12.

4.4. Comparison to SOTA methods

4.4.1. Comparison methods

To evaluate the effectiveness of the proposed polyp segmentation method, we compare it with eight SOTA methods, including UNet [4], ResUnet [5], ResUnet++[6], CFANet [26], CANet [34], PraNet [11], LW-MHFI-Net [36] and HiFormer (HiFormer-B was used in this paper) [37]. Specifically, CFANet, CANet, PraNet and HiFormer are retrained and tested using their officially released source code. For U-Net, ResUnet and ResUnet++, we adopted the reimplementation version from GitHub with most stars to investigate its results because no official PyTorch code is available. For LW-MHFI-Net, we reproduced the model based on the original paper and retrained it to investigate the results. In addition, for a fair comparison, we employed a consistent training strategy as mentioned in Section 4.3 across all datasets, and there may be some differences in the results from the original paper due to the fact that it is not the same environment as the one used in the original paper.

4.4.2. Results on Kvasir-SEG dataset

In our study, we evaluated the performance of our proposed model against SOTA models on the Kvasir-SEG dataset. As shown in Table 1, the results, demonstrating that EHANet excels in most evaluation metrics. Notably, EHANet achieves the highest mIoU and mDice scores of 0.891 and 0.940, respectively, outperforming its closest competitor, PraNet, which scores 0.871 and 0.924. This reflects the superior accuracy of EHANet in overlapping predicted and actual segmentation regions. Additionally, our method attains the lowest Asd and Hd, indicating its superior performance in object edge delineation accuracy and segmentation consistency. EHANet's S_α value of 0.948 surpasses other models, highlighting its exceptional capability to capture the structural information of segmented objects. Furthermore, EHANet's F_ω and E_φ scores are 0.951 and 0.971, respectively, underscoring its robustness in segmentation performance. Compared to HiFormer and LW-MHFI-Net, which achieve competitive results but slightly fall behind in key metrics like mIoU and mDice, EHANet demonstrates superior accuracy and consistency across all evaluation criteria. Compared to CANet, which has a higher MAE value of 0.031, EHANet boasts the lowest MAE value of 0.021, signifying its ability to minimize the absolute difference between predicted segmented pixel values and actual images, ensuring high fidelity in image analysis tasks.

Table 1
Quantitative results of evaluation metrics for EHANet in comparison with SOTA methods on the Kvasir-SEG dataset, DSB 2018 dataset, ISIC 2018 dataset and LGG dataset. The best results are shown in **Bold**.

Dataset	Method	mIoU	mDice	Asd	Hd	S_α	F_ω	E_φ	M
Kvasir-SEG	Unet	0.688	0.779	4.796	4.230	0.809	0.925	0.944	0.055
	ResUnet	0.733	0.833	5.054	5.458	0.852	0.872	0.912	0.048
	ResUnet++	0.761	0.847	4.289	5.004	0.861	0.888	0.912	0.048
	CANet	0.824	0.884	9.220	4.254	0.903	0.906	0.936	0.031
	CFANet	0.834	0.897	3.469	5.146	0.830	0.837	0.876	0.052
	PraNet	0.871	0.924	5.988	3.674	0.943	0.956	0.967	0.037
	HiFormer	0.874	0.929	4.138	**3.461**	0.939	0.932	0.938	0.038
	LW-MHFI-Net	0.830	0.899	5.561	4.247	0.902	0.930	0.942	0.039
	EHANet(ours)	**0.891**	**0.940**	**2.499**	4.000	**0.948**	**0.951**	**0.971**	**0.021**
DSB 2018	Unet	0.779	0.814	5.399	3.571	0.906	0.874	0.878	0.036
	ResUnet	0.847	0.827	3.410	**3.525**	0.928	0.908	0.888	0.026
	ResUnet++	0.786	0.877	3.511	3.634	0.922	**0.935**	0.925	0.038
	CANet	0.838	0.910	5.510	3.527	0.911	0.934	0.968	0.027
	CFANet	0.862	0.924	3.412	3.931	0.921	0.915	0.913	0.022
	PraNet	0.869	0.928	3.390	4.250	0.903	0.895	0.957	0.028
	HiFormer	0.847	0.916	3.952	3.983	0.914	0.921	0.957	0.020
	LW-MHFI-Net	0.864	0.925	5.922	3.756	0.881	0.896	0.926	0.043
	EHANet(ours)	**0.889**	**0.943**	**3.329**	3.716	**0.934**	0.929	**0.977**	**0.018**
ISIC 2018	Unet	0.727	0.794	11.030	4.358	0.897	0.914	0.936	0.035
	ResUnet	0.755	0.835	8.748	5.257	0.851	0.863	0.892	0.055
	ResUnet++	0.796	0.868	8.826	4.550	0.876	0.895	0.911	0.048
	CANet	0.831	0.898	5.965	4.023	0.896	0.920	0.932	0.045
	CFANet	0.762	0.845	4.419	4.598	0.923	0.915	**0.971**	**0.022**
	PraNet	0.844	0.909	3.390	4.250	0.903	0.895	0.965	0.028
	HiFormer	0.832	0.895	5.667	3.789	0.901	0.917	0.936	0.039
	LW-MHFI-Net	0.799	0.876	7.216	4.112	0.863	0.910	0.918	0.046
	EHANet(ours)	**0.860**	**0.929**	**3.133**	**3.249**	**0.949**	**0.925**	0.955	0.030
LGG	Unet	0.734	0.806	5.264	3.273	0.879	0.829	0.909	0.054
	ResUnet	0.730	0.806	5.172	3.277	0.875	0.833	0.911	0.058
	ResUnet++	0.756	0.833	5.490	**3.203**	0.889	0.850	0.938	0.057
	CANet	0.750	0.829	6.249	3.164	0.888	0.849	0.930	0.061
	CFANet	0.714	0.800	4.592	3.270	0.871	0.827	0.922	0.063
	PraNet	0.731	0.817	5.469	3.224	0.883	0.854	0.938	0.060
	HiFormer	0.744	0.828	5.131	4.261	0.896	0.846	0.921	0.056
	LW-MHFI-Net	0.759	0.826	6.838	4.285	0.859	0.873	0.919	0.052
	EHANet(ours)	**0.784**	**0.862**	**4.368**	3.216	**0.917**	**0.884**	**0.939**	**0.043**

Qualitative segmentation results, presented in Fig. 5, compare our model with eight SOTA methods. Overall, proposed EHANet's results more closely resemble real-world scenarios and outperform all compared methods in handling multiple challenging factors. For instance, with small and numerous polyps, EHANet accurately segments them, while other methods either miss polyps or detect only one. In cases with polyps having complex boundaries, EHANet demonstrates superior accuracy in segmentation. For large polyps, EHANet outperforms other methods by achieving more complete segmentation. Even with polyps affected by lighting, leading to unclear boundaries, EHANet achieves the highest segmentation completeness despite minor defects. Overall, these results validate EHANet's excellent performance in addressing various challenging factors in polyp segmentation.

4.4.3. Results on DSB 2018 dataset

In the context of the 2018 Data Science Bowl Challenge, our proposed EHANet model was rigorously evaluated against SOTA models to assess its performance in automated cell segmentation tasks. As detailed in Table 1, EHANet demonstrates superior performance across multiple evaluation metrics, highlighting its capability in accurately segmenting complex biological images. Specifically, EHANet achieves the highest mIoU and mDice scores of 0.889 and 0.943, respectively, surpassing its closest competitors, CFANet and PraNet, by 2.7 and 2.0 percentage points in mIoU. In addition, when compared to HiFormer and LW-MHFI-Net, EHANet showcases clear advantages in segmentation accuracy and boundary delineation. HiFormer, while competitive, records an mIoU of 0.847 and an mDice of 0.916, which are noticeably lower than EHANet's. Similarly, LW-MHFI-Net achieves an mIoU of 0.864 and an mDice of 0.925, performing well but still falling behind EHANet in critical metrics. EHANet's superior mIoU and mDice highlight its enhanced ability to capture more precise and complete segmentation regions, which is crucial for accurate biological analysis. This significant improvement underscores EHANet's enhanced ability to accurately capture regions of interest within images. In terms of boundary accuracy, EHANet records Asd and Hd values of 3.329 and 3.716, respectively, which indicate a precise approximation of the real contours of cell nuclei, essential for quantitative cell morphology analysis. While PraNet also performs well, EHANet's superior Asd and Hd scores (compared to PraNet's 3.390 and 4.250) emphasize its higher boundary delineation accuracy. Additionally, EHANet achieves the lowest MAE of 0.018 among all models, demonstrating its exceptional pixel-level accuracy and minimal segmentation error.

The qualitative comparison, illustrated in Fig. 6, further substantiates these findings. In the DSB2018 dataset, images often feature multiple small-sized targets with minimal color variation between the targets and the background. In these challenging scenarios, EHANet consistently produces more accurate prediction results than other methods. For instance, in the first row of Fig. 6, EHANet uniquely detects very small point regions in the true mask that other methods fail to identify. Moreover, when dealing with adjacent cells that create unclear boundaries, EHANet excels in separating these neighboring cells with higher accuracy. Conversely, other models, such as ResUnet++, CANet, Unet, and CFANet, struggle to effectively differentiate adjacent cells. This comprehensive performance evaluation solidifies EHANet's leading position in the field of automated cell segmentation, particularly under challenging conditions present in the DSB2018 dataset.

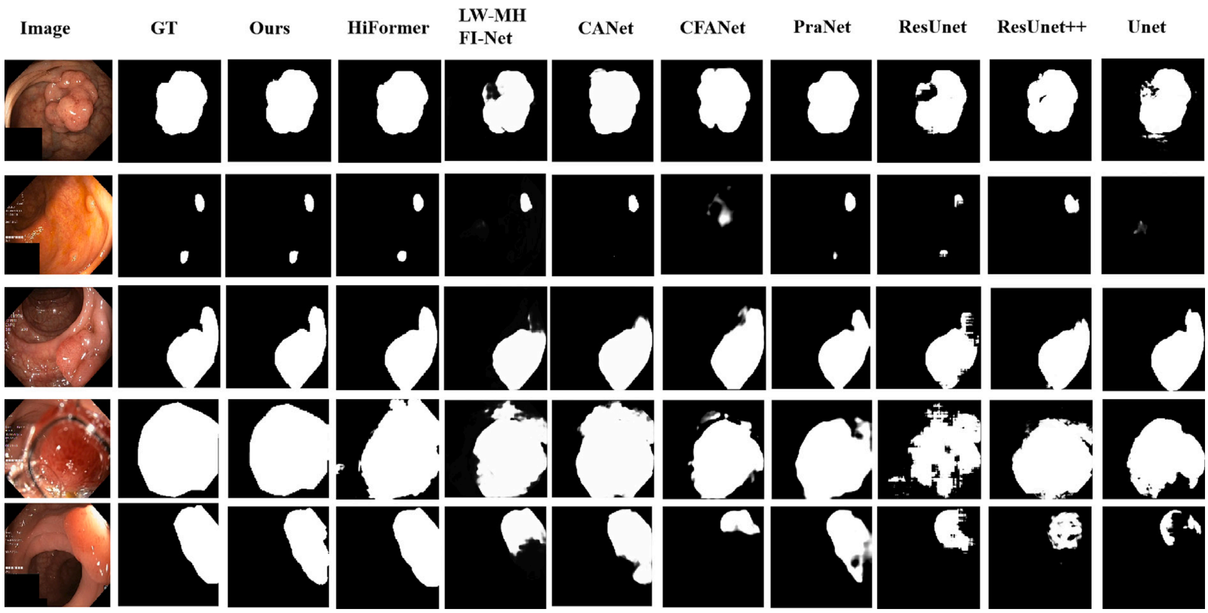

Fig. 5. Qualitative comparisons of segmentation results comparing our model with eight SOTA methods on the Kvasir-SEG dataset.

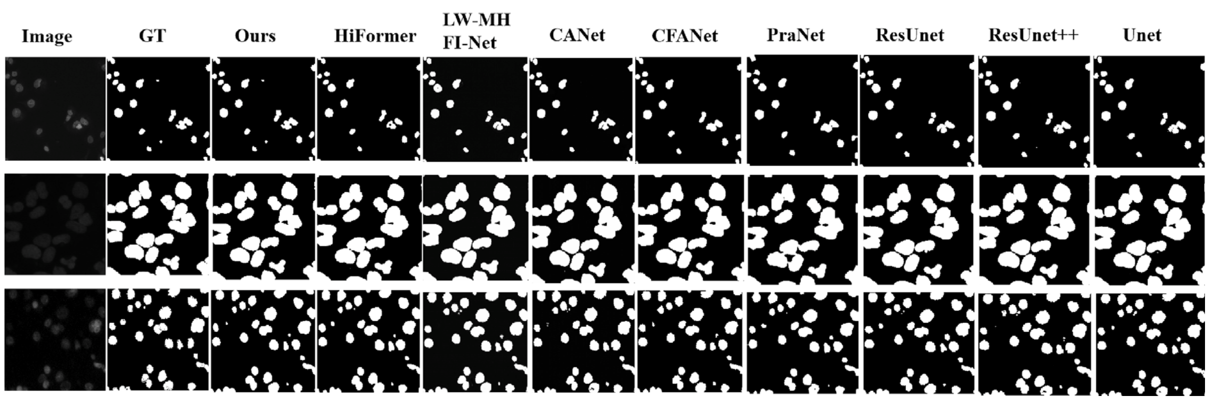

Fig. 6. Qualitative comparisons of segmentation results comparing our model with eight SOTA methods on the 2018 Data Science Bowl dataset.

4.4.4. Results on ISIC 2018 dataset

In the field of dermatological image analysis, particularly in the context of the ISIC2018 dataset, our EHANet model was benchmarked against state-of-the-art (SOTA) models to evaluate its efficacy in skin image segmentation. The quantitative results, as detailed in Table 1, demonstrate EHANet's superior performance across various evaluation metrics. Specifically, EHANet achieved the highest mIoU and mDice scores of 0.860 and 0.929, respectively, outperforming its closest competitor, PraNet, by 1.6 percentage points in mIoU. This significant improvement underscores EHANet's enhanced ability to accurately predict lesion boundaries, which is crucial for early detection and treatment planning of skin cancer. When comparing HiFormer and LW-MHFI-Net with EHANet, several additional metrics reinforce EHANet's superiority. For instance, EHANet achieved an S_α value of 0.949, outperforming both HiFormer (0.901) and LW-MHFI-Net (0.863). This indicates EHANet's superior ability to preserve the structural integrity of the lesion during segmentation, an essential factor for capturing the overall shape and layout of skin abnormalities. EHANet's Asd and Hd scores of 3.133 and 3.249, respectively, highlight its exceptional boundary delineation accuracy, crucial for accurately capturing the true contours of lesions. These capabilities are essential for assessing the severity and potential malignancy of skin lesions.

The qualitative comparison, illustrated in Fig. 7, further supports these quantitative findings. In the ISIC2018 dataset, lesion regions exhibit diverse color depths, yet the color variation between the target lesions and the background is relatively subtle. Under these challenging conditions, EHANet consistently provides more accurate prediction results compared to other methods. For instance, in the third row of Fig. 7, where the color difference between the lesion area and the surrounding skin is weak, EHANet successfully identifies the lesion area without misclassifying normal skin as lesions. Conversely, other models like CANet, CFANet, PraNet, ResUnet++, and Unet struggle with this distinction. Although ResUnet recognizes the lesion region in this scenario, its coverage is incomplete. In other cases where the color difference between the lesion region and the background is more pronounced but internal color variations and complex edges present additional challenges, EHANet demonstrates superior accuracy. Across all rows, EHANet consistently identifies the lesion areas more effectively than other methods, solidifying its role as a highly effective tool for the early detection and accurate segmentation of skin lesions. These comprehensive performance evaluations underscore EHANet's leading capability in addressing the challenges of skin image segmentation in the ISIC2018 dataset.

4.4.5. Results on LGG dataset

In an exhaustive evaluation of the low-grade glioma dataset, we compared the performance of the EHANet model with that of current SOTA models to assess its effectiveness in MRI image segmentation of

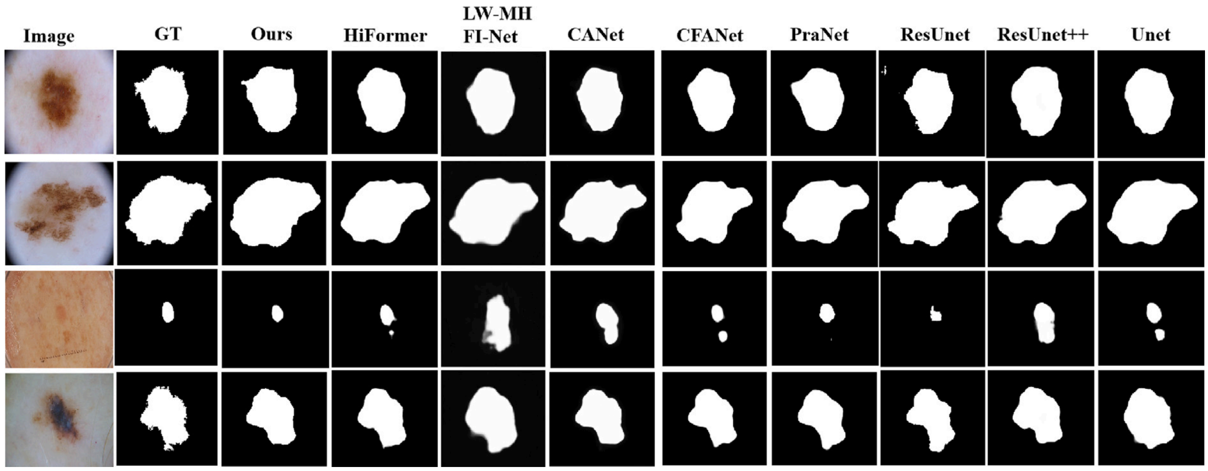

Fig. 7. Qualitative comparisons of segmentation results comparing our model with eight SOTA methods on the 2018 ISIC dataset.

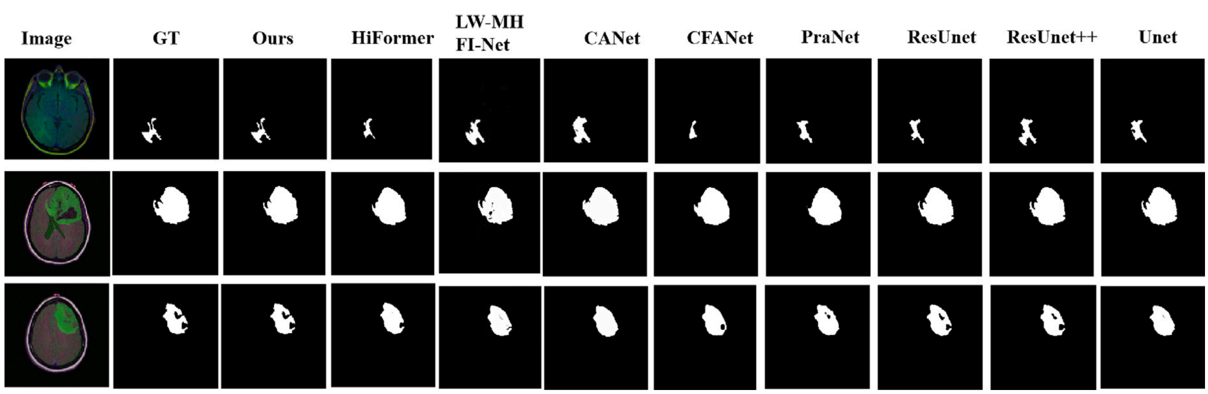

Fig. 8. Qualitative comparisons of segmentation results comparing our model with eight SOTA methods on the LGG dataset.

low-grade gliomas. The results, as shown in Table 1, demonstrate the excellent performance of EHANet across various metrics, reinforcing its applicability in improving the accuracy of medical image analysis, particularly in the field of neuro-oncology. Specifically, EHANet emerges as a new benchmark among comparative models with an mIoU of 0.784 and an mDice of 0.862. These results highlight EHANet's superior ability to accurately predict ROIs, ensuring a higher degree of overlap between the predicted mask and the ground truth mask. For example, EHANet's mIoU outperforms the previous leading ResUnet++ model, which had an mIoU score of 0.756. Additionally, EHANet's S_α of 0.917, F_ω of 0.884, and E_φ of 0.939 further demonstrate its robustness. These scores not only outperform other models but also illustrate EHANet's capability in capturing the overall structure and local details of glioma images, contributing to a comprehensive understanding of tumor characteristics.

LGG images typically exhibit complex structures and diverse features, such as varying tumor size, shape, and location from individual to individual. As shown in Fig. 8, in the first row, the color difference between the tumor and the background is not pronounced due to light variations and shooting angle, and the tumor is relatively small. Other methods fail to accurately identify the size and outline of the tumor. In contrast, the tumor area detected by EHANet closely matches the labeled real mask. Moreover, in the second and third rows, where the tumors are larger and more distinct from the background but have complex boundaries, EHANet demonstrates superior accuracy and completeness in both edge delineation and overall tumor identification compared to other methods. These comprehensive performance evaluations underscore EHANet's leading capability in addressing the challenges of MRI image segmentation in the LGG dataset, making it a highly effective tool for accurate and reliable medical image analysis in neuro-oncology.

4.4.6. Grad-CAM visualization and model interpretability analysis

To enhance the interpretability of our model in the medical image segmentation task, we applied Grad-CAM (Gradient-weighted Class Activation Mapping) across four datasets. The activation maps generated in Fig. 9 clearly demonstrate that the model's attention is predominantly focused on the lesion areas. Whether in gastrointestinal lesion detection from endoscopic images, nuclear segmentation tasks, skin lesion detection, or low-grade glioma segmentation, Grad-CAM consistently shows that the model effectively targets the lesion regions. This is especially evident in cases where boundaries are indistinct or the lesion areas are relatively small, with the activation maps still exhibiting strong robustness. The introduction of Grad-CAM enhances the transparency of the model and the basis of the model's decision-making in complex scenarios

4.4.7. Model complexity and inference time comparison

To investigate the model complexity and inference time, we report the model sizes and inference times of our model and other compared methods in Table 2. Here, parameters are measured in million (M), floating point operations (FLOPs) are measured in giga (G), and the inference time is measured by frames per second (FPS). As can be observed, our model is with the second minimal parameters in comparison with the other methods. Because our model adopts an Inter-layer Edge-aware Module to generate the edge map, it takes much more inference time for the segmentation than other compared methods. In future work, we focus on designing more lightweight networks to

Table 2
Comparison of model size and inference time.

Models	Unet	ResUnet	ResUnet++	CFANet	CANet	PraNet	HiFormer	LW-MHFI-Net	EHANet(ours)
Speed (FPS)	123.11	23.50	64.57	23.50	83.19	25.05	60.73	48.27	65.14
FLOPs (G)	123.87	55.36	70.99	55.36	25.31	13.15	32.02	49.78	38.54
Params (M)	34.52	18.22	57.93	25.24	24.12	30.50	25.51	30.24	21.96

Table 3
Quantitative ablation results of evaluation metrics for EHANet on the Kvasir-SEG dataset, DSB 2018 dataset, ISIC 2018 dataset and LGG dataset.

Dataset	Setting	mIoU	mDice	Asd	Hd	S_α	F_ω	E_φ
Kvasir-SEG	w/o EFA	0.876	0.913	3.343	4.772	0.927	0.923	0.947
	w/o AHFA	0.866	0.927	4.130	5.024	0.922	0.941	0.960
	w/o IEM	0.868	0.928	6.846	5.781	0.941	0.947	0.958
	EHANet(ours)	0.891	0.940	2.499	4.000	0.948	0.951	0.971
DSB 2018	w/o EFA	0.865	0.926	3.543	3.772	0.923	0.925	0.965
	w/o AHFA	0.855	0.915	3.388	3.813	0.928	0.920	0.972
	w/o IEM	0.871	0.904	4.381	3.752	0.932	0.926	0.971
	EHANet(ours)	0.889	0.943	3.329	3.716	0.934	0.929	0.977
ISIC 2018	w/o EFA	0.825	0.909	4.747	4.551	0.937	0.915	0.914
	w/o AHFA	0.828	0.910	3.932	4.693	0.937	0.914	0.923
	w/o IEM	0.815	0.904	5.835	4.832	0.925	0.918	0.941
	EHANet(ours)	0.860	0.929	3.133	3.249	0.949	0.925	0.955
LGG	w/o EFA	0.760	0.841	5.459	3.971	0.893	0.845	0.927
	w/o AHFA	0.745	0.824	5.372	3.567	0.893	0.851	0.931
	w/o IEM	0.744	0.825	5.955	4.73	0.890	0.844	0.934
	EHANet(ours)	0.784	0.862	4.368	3.216	0.917	0.884	0.939

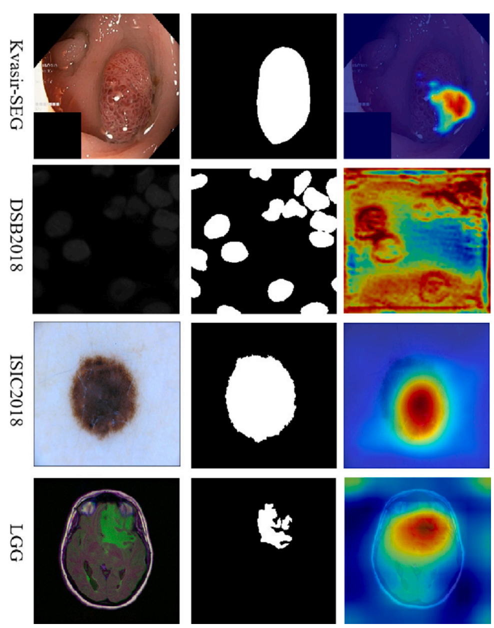

Fig. 9. Visualization of Grad-CAM on four different medical datasets.

improve the efficiency of the proposed model for real-time medical image segmentation.

4.5. Ablation study

This section investigates the importance of three key components in the proposed EHANet. The ablative results are shown in Table 3.

(1) Effectiveness of EFA: In order to verify the importance of EFA, we replace it with the original jump connection called "w/o EFA". As shown in Table 3, in the absence of EFA (w/o EFA), as shown in the Kvasir-SEG dataset, the mIoU of the model decreases from 0.891 to 0.876 in EHANet, and the mDice decreases as well, indicating that the accuracy of the image segmentation is affected. Meanwhile, in the Kvasir-SEG dataset, S_α decreased from 0.948 to 0.927, showing that the model is less sensitive to details in the segmentation task. In addition, F_ω decreased from 0.951 to 0.923, which demonstrated a combined decrease in the accuracy and recall of the segmentation results. E_φ also decreased, proving a weakening of the overall segmentation effect. The changes in these metrics further validate the important role of EFA in enhancing the model's ability to capture medical image details. Combined with the comparative plots in Fig. 10, it can be seen that the model lacking EFA lacks segmentation details in its prediction. These changes confirm the important role of EFA in enhancing the model's ability to capture details, which can improve performance in image analysis and understanding tasks by enhancing the model's ability to learn channel attention and spatial attention.

(2) Effectiveness of AHFA: To investigate the contribution of AHFA, we employed two 3 × 3 convolutional layers with batch normalization and ReLU activation function as decoding blocks to replace AHFA, referred to as "w/o AHFA". In the Kvasir-SEG dataset, after removing the AHFA, the mIoU and mDice drop to 0.866 and 0.927, respectively, showing a significant degradation of the model performance. In particular, Hd increases from 4.000 to 5.024, indicating greater inaccuracy in the segmentation boundaries under worst-case scenarios. Similarly, other metrics such as S_α, F_ω, and E_φ exhibit a decreasing trend, with F_ω reducing from 0.951 to 0.941, reflecting a decline in the balance of accuracy and recall. Fig. 10 clearly shows the large difference between the model predictions without AFHA blocks and the ground truth. The main reason for this is that the overall prediction of the model will not be able to accurately understand the complex structures and relationships in the image due to the loss of contextual information and insufficient feature fusion. Meanwhile, the observations suggest that AFHA plays a critical role in maintaining high segmentation performance across various datasets by effectively fusing cross-layer features and integrating local and global contextual information.

(3) Effectiveness of IEM: To investigate the contribution of IEM, we remove it from the proposed network, namely "w/o IEM". Without using IEM (w/o IEM), as in the DSB 2018 dataset, the mIoU and mDice of the model are 0.871 and 0.904, respectively, which are decreased compared to EHANet. More importantly, the significant increase in Asd and Hd on the four datasets indicates an increase in the inconsistency between the segmentation results and the actual boundaries. For example, on the Kvasir-SEG and LGG datasets in Fig. 10, the shape of the mask map predicted by the model missing IEM is very different from the actual one, which is in large part due to the missing information known about the boundary. This shows the importance of IEM in improving the accuracy of boundary prediction and the quality of segmentation.

4.6. Discuss

This study introduced the Edge-guided and Hierarchical Aggregation Network (EHANet) to tackle challenges in medical image segmentation, such as blurred boundaries, complex lesion structures, and varying scales of abnormalities. EHANet integrates novel modules like the Inter-layer Edge-aware Module (IEM) and Efficient Fusion Attention (EFA) Module, which enhance its ability to capture fine spatial details and maintain accurate segmentation boundaries. Extensive experiments on four public datasets (Kvasir-SEG, DSB 2018, ISIC 2018, and LGG) demonstrated that EHANet consistently outperforms state-of-the-art models, showing superior segmentation accuracy and robustness.

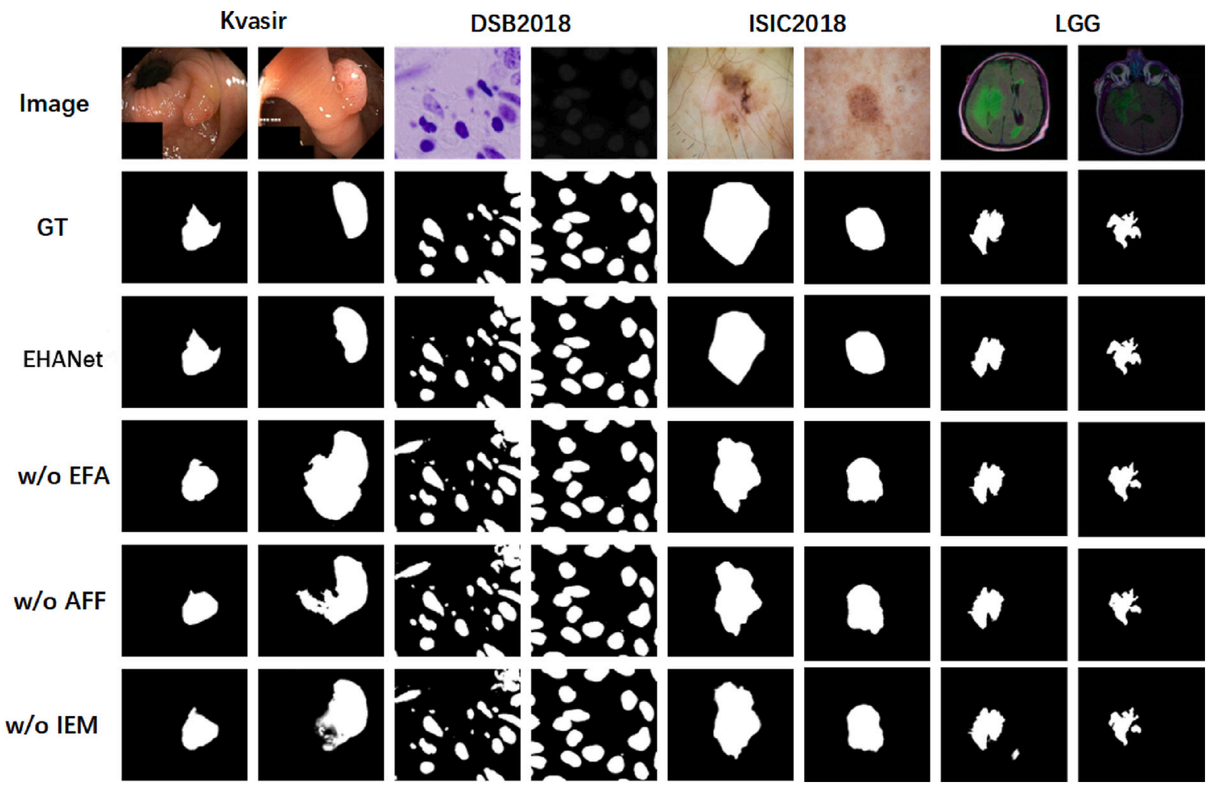

Fig. 10. Qualitative comparisons of segmentation results comparing our model with three ablation studies on four datasets. Here, GT represent ground truth mask.

In comparison to existing models, such as UNet and its variants (ResUnet, ResUnet++), these models struggle with capturing complex contextual information due to their simple architectures and lack of multi-scale fusion. This limitation makes them less effective at segmenting intricate structures or lesions with irregular boundaries. Models like PraNet, CANet, CFANet, LW-MHFI-Net and HiFormer address this through multi-scale feature fusion, but still encounter difficulties in boundary delineation, particularly with complex lesion shapes or textures. Their reliance on multi-scale fusion also introduces higher computational overhead, and the absence of explicit edge guidance can result in less precise segmentation in challenging scenarios. EHANet overcomes these limitations by incorporating IEM, enhancing boundary detection while maintaining computational efficiency, ensuring accurate segmentation across varying scales and complex cases. Additionally, for HiFormer, which employs a hybrid transformer-based module to capture long-range dependencies, EHANet addresses its limitations, such as the high number of parameters that increase computational demands. By utilizing a pre-trained transformer backbone that is frozen during training, EHANet efficiently captures global information while reducing trainable parameters, leading to faster training and lower computational costs.

However, despite its success, EHANet has certain limitations. Its reliance on edge-guided mechanisms may reduce its performance in cases where edge information is weak or obscured by noise. Additionally, the transformer-based approach increases computational complexity, which may limit its application in real-time or resource-constrained environments. The current study also focused primarily on 2D datasets, and further validation is needed for 3D medical image segmentation tasks, such as those involving MRI or CT scans.

To address these limitations, future work could explore advanced preprocessing techniques, such as noise reduction, contrast enhancement, and adaptive thresholding, to improve image quality before feeding it into the model. Incorporating multimodal data from different imaging techniques (e.g., CT, MRI) could also improve the model's ability to handle diverse image types and provide more comprehensive segmentation results. For reducing computational complexity, techniques such as model pruning, quantization, and knowledge distillation can be employed to optimize EHANet for real-time applications. Extending EHANet to 3D segmentation by incorporating volumetric data processing with 3D convolutions and attention mechanisms would enhance its utility in tasks involving more complex imaging modalities.

5. Conclusion

In this study a novel Edge-guided and Hierarchical Aggregation Network was proposed. This network was designed to improve the accuracy and efficiency of medical image segmentation. EHANet combines IEM and an efficient EFA module, which significantly enhances the boundary detection capability and optimizes information utilization. IEM significantly enhances the edge detection capability and optimizes the information utilization by exploiting multiscale decoding layers in the information transfer and fusion mechanisms, it strengthens the expression of edge details and is able to accurately capture subtle edge changes between lesion areas and normal tissues. It has a direct impact on improving the accuracy of clinical diagnosis. Meanwhile, the EFA module reduces information redundancy by optimizing channel and spatial information utilization and focuses on the key regions of the image, which improves the accuracy of segmentation and the adaptability of the model to the background of complex medical images. In addition, the use of the AHFA as decoders in conjunction with the IEM significantly improves the handling of scale variations and the overall segmentation performance. This cross-layer feature fusion strategy provides the model with the ability to handle information at different scales from microscopic details to macroscopic structures, which is especially critical when dealing with medical images as they usually contain structural information at multiple scales from cellular level to organ level. Experimental results demonstrate that EHANet shows significant advantages over existing SOTA methods on multiple medical image datasets.

In future studies, it is planned to explore the application of EHANet to more complex 3D image data, such as CT and MRI scans. Also, it is planned to further enhance the robustness of the network in dealing with image noise and artifacts.

CRediT authorship contribution statement

Yi Tang: Writing – original draft, Visualization, Software, Methodology, Conceptualization. **Di Zhao:** Writing – original draft, Formal analysis, Conceptualization. **Dmitry Pertsau:** Writing – review & editing, Supervision. **Alevtina Gourinovitch:** Writing – review & editing, Supervision. **Dziana Kupryianava:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The source code is available at: https://github.com/tyjcbzd/EHANet.

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